

ELECTRIC CIRCUITS

PART 4 POWER, MAINS AND THE GRID

LESSONS 11–12

Power and energy, mains safety and the National Grid, then the unit review. Learn with it in class and revise from it at home.

11. Power and energy transfer

12. Mains, safety and the National Grid

Unit review · Static electricity (Physics only)

ANSWER EDITION · FOR TEACHERS AND SELF-MARKING



NAME

CLASS

TEACHER

TARGET GRADE

HOW THIS BOOKLET WORKS

This is **Part 4** of the Electric Circuits workbook. It builds on the parts before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

 DO NOW Quick questions from earlier lessons. Remembering again is what makes it stick.	 LEARN IT The explanation, diagrams and worked examples. Read this again when you revise.	 TRY IT Practice that builds up: first with support, then on your own.
 LAB Record measurements from a virtual lab or a required practical.	 EXAM CORNER An AQA-style question with a tip on how the marks are given.	 REVISE IT Summary, cover-and-answer questions and an “I can” check to colour in.

 **Revising at home?** For each lesson: (1) read the Learn it pages, (2) cover them and answer the “Cover the page” questions in Revise it, (3) check with the quick answers at the back, (4) do the matching mission on the companion website, which has more questions with answers.

THE ROUTE THROUGH THE UNIT · THIS PART IN BOLD

#	LESSON	COMPANION WEBSITE
01	Complete circuits and symbols What a circuit needs, circuit symbols and drawing rules, measuring current	Mission 1 · Build & measure
02	Current and charge Current as a rate of flow of charge, $Q = I t$, current is not used up	Mission 2 · Charge on the move
03	Potential difference p.d. as energy per coulomb, voltmeters, $E = Q V$, p.d.s add in series	Mission 3 · Energy per coulomb
04	Resistance and $V = I R$ Resistance, $V = I R$, ohmic conductors, why heating raises resistance	Mission 4 · Resistance & $V = I R$
05	Resistance of a wire REQUIRED PRACTICAL Required practical: how the length of a wire affects its resistance	Required practical · Resistance
06	Series and parallel circuits Current, p.d. and resistance rules for series and parallel circuits	Mission 5 · Series & parallel
07	Resistors in series and parallel REQUIRED PRACTICAL Required practical: measure total resistance of two resistors in series and in parallel	Required practical · Resistance (Part B)
08	I–V graphs I–V graphs of a fixed resistor, filament lamp and diode	Mission 6 · I–V characteristics
09	Investigating I–V characteristics REQUIRED PRACTICAL Required practical: measure the I–V characteristics of a resistor, lamp and diode	Required practical · I–V characteristics
10	Thermistors, LDRs and sensors How thermistors and LDRs respond to temperature and light, and sensor circuits	Mission 6 · Thermistor and LDR
11	Power and energy transfer Power, $E = P t$, $P = V I$, $P = I^2 R$, energy transfers in appliances	Mission 7 · Power & the National Grid
12	Mains, safety and the National Grid ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid	Missions 7 and 8

Companion website: panphy.app/gcsephy/year10phy/unit01 · missions, interactive equation triangles, required-practical pages and the Exam Zone.

EQUATION TOOLKIT

You will meet these one at a time. Learn each equation **in words** as well as symbols: your exams may or may not give you an equation sheet, and knowing them by heart makes every question faster.

$$Q = I t$$

Lesson 2

charge flow = current × time

C, A, s

Rearranged: $I = Q \div t$ · $t = Q \div I$

$$E = Q V$$

Lesson 3

energy transferred = charge × p.d.

J, C, V

Rearranged: $V = E \div Q$ · $Q = E \div V$

$$V = I R$$

Lesson 4

p.d. = current × resistance

V, A, Ω

Rearranged: $I = V \div R$ · $R = V \div I$

$$R_{\text{total}} = R_1 + R_2$$

Lesson 6

total resistance in series

Ω

Rearranged: (series only)

$$P = V I$$

Lesson 11

power = p.d. × current

W, V, A

Rearranged: $V = P \div I$ · $I = P \div V$

$$P = I^2 R$$

Lesson 11

power = current² × resistance

W, A, Ω

Rearranged: $R = P \div I^2$ · $I = \sqrt{(P \div R)}$

$$E = P t$$

Lesson 11

energy transferred = power × time

J, W, s

Rearranged: $P = E \div t$ · $t = E \div P$

$$V_p I_p = V_s I_s$$

Lesson 12

transformer: power in = power out
(Higher tier)

V, A

Rearranged: $I_s = V_p I_p \div V_s$

FOUR STEPS FOR EVERY CALCULATION

1

Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2

Write the equation you will use, in symbols.

3

Substitute the numbers, then rearrange if needed.

4

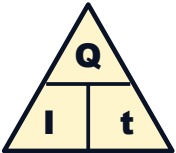
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes → seconds	× 60	3.0 min = 180 s
hours → seconds	× 3600	2.0 h = 7200 s
milliamps (mA) → amps (A)	÷ 1000	250 mA = 0.25 A
kilowatts (kW) → watts (W) · kilohms (k Ω) → ohms (Ω)	× 1000	2.2 kW = 2200 W
kilovolts (kV) → volts (V)	× 1000	400 kV = 400 000 V

■ EQUATION TRIANGLES: HELP WITH REARRANGING

- 1 Cover the quantity you want to find.
- 2 The two left **side by side**? **Multiply** them.
- 3 One **above** the other? **Divide** the top by the bottom.

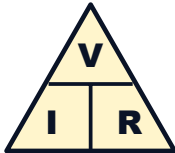
Example: to find I from $Q = I t$, cover I . Q is above t , so $I = Q \div t$. Triangles only help with rearranging: you still need to learn each equation. For $P = I^2 R$, the triangle gives I^2 ; take the square root to find I .



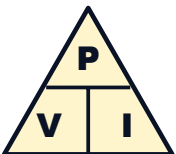
$Q = I t$
 $Q = I \times t$
 $I = Q \div t$
 $t = Q \div I$



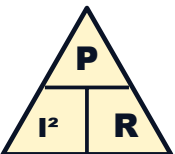
$E = Q V$
 $E = Q \times V$
 $Q = E \div V$
 $V = E \div Q$



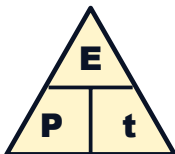
$V = I R$
 $V = I \times R$
 $I = V \div R$
 $R = V \div I$



$P = V I$
 $P = V \times I$
 $V = P \div I$
 $I = P \div V$



$P = I^2 R$
 $P = I^2 \times R$
 $R = P \div I^2$
 $I^2 = P \div R$, so $I = \sqrt{P \div R}$



$E = P t$
 $E = P \times t$
 $P = E \div t$
 $t = E \div P$

Interactive version: panphy.app/gcsephy/year10phy/unit01/equation-triangles.

POWER AND ENERGY TRANSFER

AQA Combined 6.2.4.1–6.2.4.2 · Physics 4.2.4.1–4.2.4.2

BIG QUESTION Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?

BY THE END YOU CAN

- describe power as the energy transferred per second
- use $E = P t$, $P = V I$ and $P = I^2 R$ with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

power

watt (W)

kilowatt (kW)

energy transferred

power rating

ONLINE COMPANION

Mission 7 · Power & the National Grid

Read the power and energy sections, then try questions 1–4.

panphy.app/gcsephy/year10phy/unit01/lesson/power-and-national-grid/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

5.0 C passes through a 12 V lamp.
How much energy is transferred?

$$E = Q V = 60 \text{ J}$$

2

2.0 A flows for 1 minute. How much
charge flows?

$$Q = 2.0 \times 60 = 120 \text{ C}$$

3

What is the resistance of a heater
with 230 V across it and 5.0 A
through it?

$$R = 230 \div 5.0 = 46 \text{ } \Omega$$



EXPLAIN

HOW FAST IS ENERGY TRANSFERRED?

Power is the **energy transferred per second**. It is measured in **watts (W)**: 1 W = 1 J per second. A 2000 W (2 kW) kettle transfers 2000 J of energy every second.

APPLIANCE	TYPICAL POWER	ENERGY TRANSFERRED EACH SECOND
LED lamp	10 W	10 J
Laptop	60 W	60 J
Hairdryer	1.2 kW = 1200 W	1200 J
Kettle	2.2 kW = 2200 W	2200 J
Electric shower	9 kW = 9000 W	9000 J

Every appliance transfers energy electrically from the mains to other energy stores: a kettle to the **thermal store** of the water; a hairdryer's fan to **kinetic** energy and its heater to thermal; a phone charger to the **chemical store** of the battery.

$$E = P t$$

energy transferred = power × time

E in joules (J) · P in watts (W) · t in seconds (s)

Convert kW → W (× 1000) and minutes or hours → seconds first.



WORKED EXAMPLE

A 2.0 kW kettle is on for 3.0 minutes. Calculate the energy transferred.

KNOW $P = 2000 \text{ W}$, $t = 180 \text{ s}$

EQUATION $E = P t$

SUBSTITUTE $E = 2000 \times 180$

ANSWER $E = 360\,000 \text{ J}$

YOUR TURN

A 1.2 kW hairdryer is used for 10 minutes. Calculate the energy transferred.

KNOW $P = 1200 \text{ W}$, $T = 600 \text{ S}$

EQUATION $E = P T$

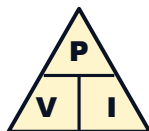
SUBSTITUTE $E = 1200 \times 600$

ANSWER $E = 720\,000 \text{ J}$

$$P = V I$$

power = p.d. ×
current

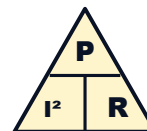
P in W · V in V · I
in A



$$P = I^2 R$$

power =
current² ×
resistance

P in W · I in A · R
in Ω



WORKED EXAMPLE

A heater on the 230 V mains draws a current of 4.0 A. Calculate its power.

EQUATION $P = V I$

SUBSTITUTE $P = 230 \times 4.0$

ANSWER $P = 920 \text{ W}$

YOUR TURN

A current of 3.0 A flows through a 20 Ω resistor. Calculate the power.

EQUATION $P = I^2 R$

SUBSTITUTE $P = 3.0^2 \times 20 = 9.0 \times 20$

ANSWER $P = 180 \text{ W}$

! In $P = I^2 R$, **square the current first**. Doubling the current makes the heating **four times** bigger: that is why cables carrying big currents get hot.

IF YOU KNOW...	USE
p.d. and current	$P = V I$
current and resistance	$P = I^2 R$
power and time	$E = P t$
charge and p.d.	$E = Q V$
current and time	$Q = I t$



PRACTISE
TRY IT

- 1 Answer the big question. Show both calculations.

Kettle: $E = 3000 \times 180 = 540\,000\text{ J}$. LED: $t = 5 \times 3600 = 18\,000\text{ s}$, $E = 10 \times 18\,000 = 180\,000\text{ J}$. The kettle transfers 3 times as much energy.

- 2 **SUPPORTED** A 12 V car headlamp draws a current of 5.0 A. Calculate its power.

EQUATION	$P = VI$	
SUBSTITUTE	$P = 12 \times 5.0$	
ANSWER	60 W	unit:

- 3 **SUPPORTED** A 9.0 kW electric shower runs for 8.0 minutes. Calculate the energy transferred in joules.

CONVERT	$P = 9000\text{ W}$, $t = 480\text{ s}$	
EQUATION	$E = Pt$	
SUBSTITUTE	$E = 9000 \times 480$	
ANSWER	$4\,320\,000\text{ J}$	unit:

- 4 **ON YOUR OWN** A current of 2.0 A flows through a $10\ \Omega$ resistor. Then the current is increased to 4.0 A. Calculate the power each time. What happens to the power when the current doubles?

$P = 2.0^2 \times 10 = 40\text{ W}$; $P = 4.0^2 \times 10 = 160\text{ W}$. The power is 4 times bigger.

- 5 **ON YOUR OWN** A device transfers 240 J of energy when 40 C of charge flows through it. Calculate the p.d. across it.

$$V = E \div Q = 240 \div 40 = 6.0\text{ V}$$

- 6 **CHALLENGE** A lamp is rated 60 W, 230 V. Calculate the current through it and its resistance.

$$I = P \div V = 60 \div 230 = 0.26\text{ A}. R = V \div I = 230 \div 0.26 \approx 880\ \Omega.$$



Get the marks: Write the power in watts before you substitute: $2.2 \text{ kW} = 2200 \text{ W}$. Give answers to a sensible number of significant figures.

- 7** A 2.2 kW kettle runs on the 230 V mains. (a) Calculate the current through the kettle. (b) Calculate the energy transferred in 150 s . (c) A heater draws twice this current through the same resistance. How does its heating power compare? **[6 marks]**

(a) $P = 2200 \text{ W}$ (1); $I = P \div V = 2200 \div 230$ (1) = 9.6 A (1). (b) $E = P t = 2200 \times 150$ (1) = $330\,000 \text{ J}$ (1). (c) $P = I^2 R$, so it is 4 times the power (1).

**SUMMARY**

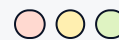
- Power is energy transferred per second. $1 \text{ W} = 1 \text{ J/s}$.
- $E = P t$ (P in W , t in s).
- $P = V I$ and $P = I^2 R$.
- Doubling the current through a resistor makes the heating 4 times bigger.
- Appliances transfer energy electrically to thermal, kinetic or chemical stores.

COVER THE PAGE AND ANSWER

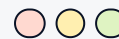
- | | |
|--|---|
| 1 What is power?
..... | 2 Convert 3.5 kW into watts.
..... |
| 3 Write two equations for electrical power.
..... | 4 Write the equation linking energy, power and time.
..... |

I CAN...

describe power as energy transferred per second



use $E = P t$ with unit conversions



use $P = V I$ and $P = I^2 R$



choose the right equation from the data given



MAINS, SAFETY AND THE NATIONAL GRID

AQA Combined 6.2.3, 6.2.4.3 · Physics 4.2.3, 4.2.4.3

BIG QUESTION Pylon cables carry up to 400 000 V. Why so high, and why do we only get 230 V at the plug?

BY THE END YOU CAN

- describe the difference between ac and dc
- state the UK mains values: 230 V, 50 Hz
- explain how the earth wire and fuse protect you
- explain why the National Grid uses very high p.d.s

alternating (ac)

direct (dc)

live

neutral

earth

fuse

National Grid

transformer

ONLINE COMPANION

Missions 7 and 8

Mission 8: Mains & safety, then the National Grid sections of Mission 7.

panphy.app/gcsephy/year10phy/unit01/lesson/mains-electricity/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

A 230 V appliance draws 10 A.
Calculate its power.

$P = VI = 2300 \text{ W}$

2

The current in a cable is halved.
What happens to the heating ($P = I^2 R$)?

It falls to a quarter.

3

What is the unit of power?

Watt (W).



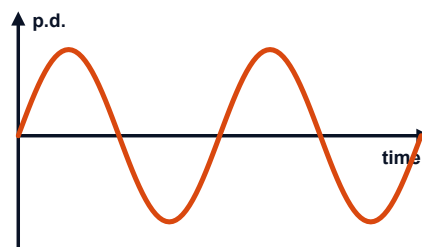
EXPLAIN

FROM POWER STATION TO PLUG

DIRECT AND ALTERNATING P.D.



dc from a cell or battery: the p.d. stays in one direction, so the current flows one way.



ac from the mains: the p.d. keeps reversing, so the current keeps changing direction.



UK mains: alternating current (ac), about 230 V, frequency 50 Hz (50 complete cycles every second).

INSIDE A PLUG CABLE

WIRE	COLOUR	JOB
Live	brown	carries the alternating p.d. from the supply (about 230 V)
Neutral	blue	completes the circuit; at or close to 0 V
Earth	green and yellow	safety wire at 0 V; only carries a current if there is a fault



● Live · brown ● Neutral · blue
● Earth · green & yellow

The outer sheath holds all three insulated wires together.

WHY THE LIVE WIRE IS DANGEROUS

Your body is at 0 V (the same as the earth). The live wire is at about 230 V. If you touch it, there is a p.d. across your body, so a current flows through you to earth: an **electric shock**, which can kill. This can happen even when the appliance is switched off, because the live wire is still connected to the supply.

HOW THE EARTH WIRE AND FUSE PROTECT YOU

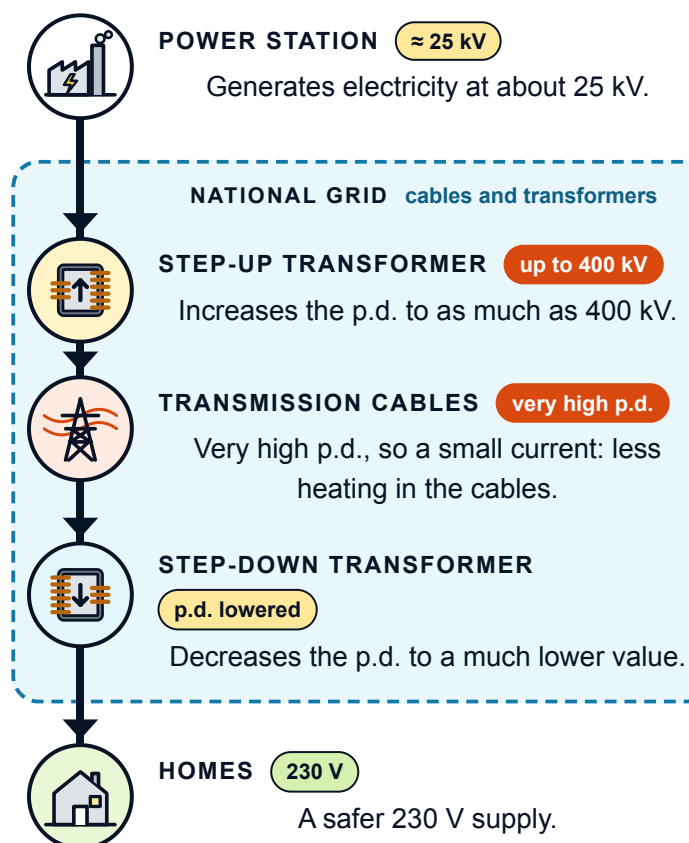
- 1 A fault makes the live wire touch the metal case.
- 2 The earth wire connects the case to earth through a very low resistance.
- 3 A very large current flows from the live wire, through the case and earth wire, to earth.
- 4 The large current melts the fuse (or trips the circuit breaker).
- 5 The live wire is disconnected, so the case is safe to touch.

THE NATIONAL GRID

The **National Grid** is a nationwide system of cables and transformers linking power stations to homes and businesses.

Step-up transformers raise the p.d. to as much as 400 kV. For the same power ($P = VI$), a higher p.d. means a **smaller current**. A smaller current means much less heating in the cables ($P = I^2 R$), so less energy is wasted: the grid is **efficient**.

Step-down transformers then lower the p.d. to 230 V, which is safer for homes.



$$V_p I_p = V_s I_s$$

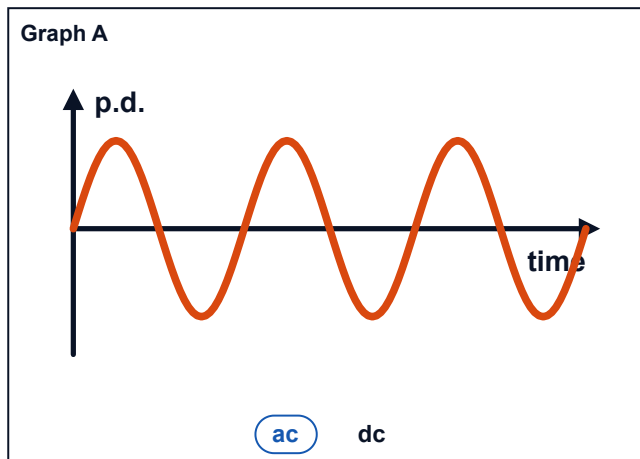
Higher tier only: for an ideal transformer, power in = power out

p = primary (input) coil · s = secondary (output) coil



PRACTISE
TRY IT

- 1 Which graph shows ac and which shows dc? Circle one for each.



- 2 Complete the sentences.

UK mains electricity is alternating with a p.d. of about 230 V and a frequency of 50 Hz. The live wire is brown, the neutral wire is blue and the earth wire is green and yellow.

WORD BANK alternating direct 230 50 live neutral earth

- 3 Number the events 1–5 to show how an earthed metal case protects you.

3 | A large current flows through the earth wire

1 | The live wire touches the metal case

5 | The case is no longer live, so it is safe to touch

2 | The case is connected to earth by a low-resistance wire

4 | The fuse melts and breaks the circuit

- 4 Explain why Ava is wrong.

Ava

Touching the live wire is only dangerous when the appliance is switched on.

The live wire is still connected to the supply at about 230 V when the switch is off. Your body is at 0 V, so there is a p.d. across you and a current flows through you to earth: a shock.

- 5 A power station sends 1 000 000 W through the grid. Use $P = VI$ to find the current at 10 000 V and at 100 000 V. Then explain why the higher p.d. is better.

$I = P \div V = 1\,000\,000 \div 10\,000 = 100\text{ A}$; $1\,000\,000 \div 100\,000 = 10\text{ A}$. The current is 10 times smaller, so the heating in the cables ($I^2 R$) is 100 times smaller: much less energy is wasted.

- 6 **HIGHER TIER** An ideal transformer has 20 kV across the primary coil and 500 A in it. The p.d. across the secondary coil is 200 kV. Calculate the current in the secondary coil.

$$V_p I_p = V_s I_s: I_s = (20\,000 \times 500) \div 200\,000 = 50\text{ A}$$



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Safety answers are sequences. Use cause $\rightarrow$ effect words: 'so', 'which means', 'therefore'.

- 7 A toaster has a metal case connected to the earth wire. A fault makes the live wire touch the case. Explain how the earth wire and the fuse protect someone who then touches the toaster. **[5 marks]**

The earth wire gives a low-resistance path from the case to earth (1). A very large current flows through the live and earth wires (1). This melts the fuse (1), which disconnects the live wire (1), so the case is not at a high p.d. and no current flows through the person (1).

- 8 Explain why the National Grid transmits electricity at a very high p.d., and why the p.d. is reduced before it reaches homes. **[4 marks]**

A step-up transformer increases the p.d. (1); for the same power, the current is smaller (1); a smaller current means less heating/energy loss in the cables, so it is more efficient (1); a step-down transformer reduces the p.d. to 230 V so it is safer in homes (1).



SUMMARY

- dc flows one way; ac keeps changing direction.
- UK mains: ac, 230 V, 50 Hz.
- Live (brown) ~230 V; neutral (blue) ~0 V; earth (green and yellow) safety wire.
- Earth wire + fuse: fault → big current → fuse melts → live disconnected.
- Grid: step up to high p.d. → small current → less heating ($I^2 R$) → efficient; step down to 230 V for safety.

COVER THE PAGE AND ANSWER

- | | |
|---|---|
| 1 State the p.d. and frequency of UK mains.
..... | 2 What colour is the live wire?
..... |
| 3 When does the earth wire carry a current?
..... | 4 Why is a high p.d. used in the National Grid?
..... |

I CAN...

describe the difference between ac and dc ○ ○ ○

state the UK mains p.d. and frequency ○ ○ ○

name the three wires and their colours and jobs ○ ○ ○

explain how the earth wire and fuse protect a user ○ ○ ○

explain why the grid uses step-up and step-down transformers ○ ○ ○

UNIT REVIEW

Mixed questions from the whole unit, like a real exam. Try each question without looking back. Then mark it with your teacher's answers and fill in the "fix it" log.



RETRIEVE

EQUATION RECALL CHALLENGE

Cover the toolkit page first.

- 1 Write each equation in symbols. Then give the unit of every quantity in it.

Equation in words	In symbols	Units
charge flow = current × time	$Q = I t$	C, A, s
energy transferred = charge × p.d.	$E = Q V$	J, C, V
p.d. = current × resistance	$V = I R$	V, A, Ω
total resistance in series	$R_{\text{total}} = R_1 + R_2$	Ω
power = p.d. × current	$P = V I$	W, V, A
power = current ² × resistance	$P = I^2 R$	W, A, Ω
energy transferred = power × time	$E = P t$	J, W, s



AQA-STYLE

MIXED EXAM PRACTICE

41 marks · about 50 minutes

- 2 A current of 0.80 A flows through a lamp for 2.5 minutes. Calculate the charge that flows. **[3 marks]**

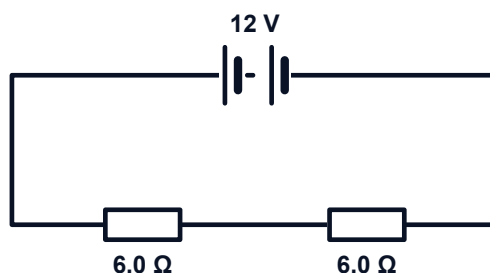
$$t = 2.5 \times 60 = 150 \text{ s (1); } Q = I t = 0.80 \times 150 \text{ (1)} = 120 \text{ C (1)}$$

- 3 A component transfers 48 J of energy when 8.0 C of charge passes through it. (a) Calculate the p.d. **[3 marks]** across it. (b) State what this p.d. means.

$$(a) V = E \div Q = 48 \div 8.0 \text{ (1)} = 6.0 \text{ V (1). (b) 6.0 J is transferred per coulomb of charge (1).}$$

- 4 Calculate the total resistance, the current, and the p.d. across one resistor.

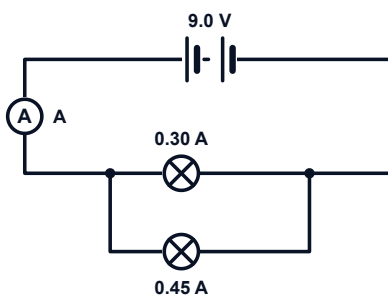
[4 marks]



$R = 6.0 + 6.0 = 12\ \Omega$ (1); $I = V \div R = 12 \div 12$ (1) $= 1.0\ \text{A}$ (1); p.d. $= 1.0 \times 6.0 = 6.0\ \text{V}$ (1)

- 5 The lamps are in parallel. State the p.d. across each lamp and the reading on ammeter A. Give reasons.

[3 marks]



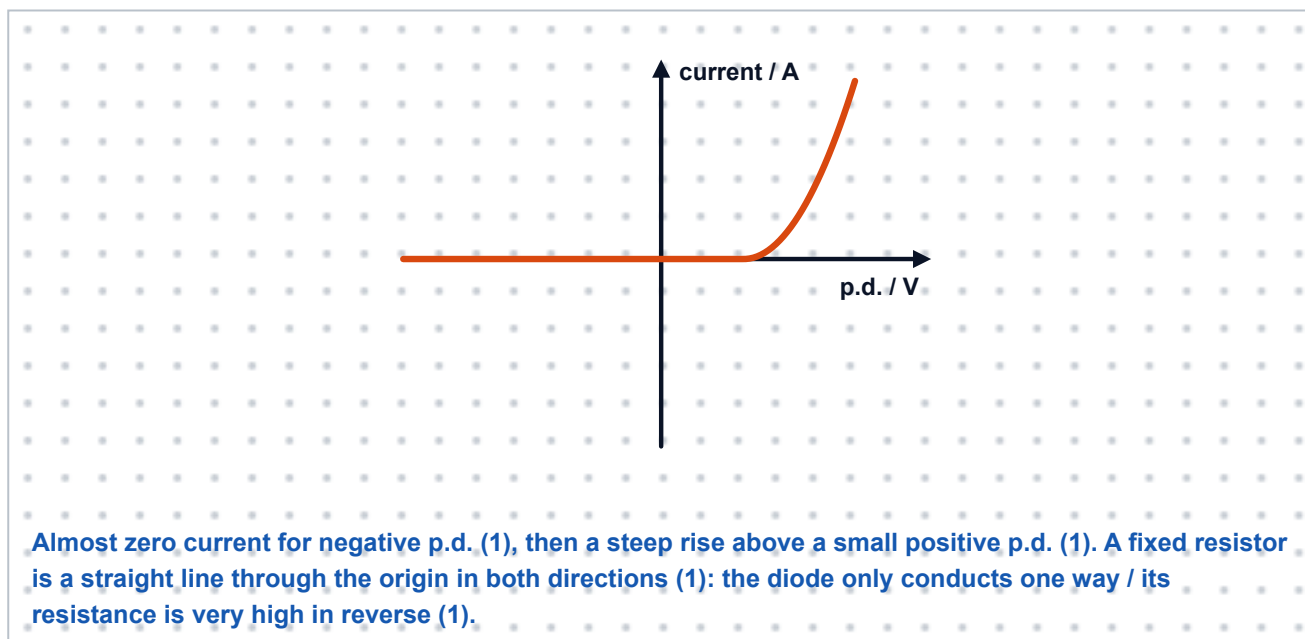
9.0 V across each lamp: in parallel each branch has the supply p.d. (1). $A = 0.30 + 0.45 = 0.75\ \text{A}$ (1), because branch currents add up to the supply current (1).

- 6 In the wire-length practical, one point on the graph of resistance against length is well above the line. (a) Suggest two possible causes. (b) What should the student do?

[4 marks]

(a) Any two: the wire heated up; poor contact at a clip; the length or a meter was misread (2). (b) Repeat that reading (1) and leave it out of the line of best fit if it is still anomalous (1).

- 7 Sketch the I–V graph of a diode. Explain how it differs from the graph of a fixed resistor. [4 marks]



- 8 A thermistor is connected in series with an ammeter and a battery. Explain how the ammeter reading changes as the thermistor cools down. [3 marks]

As it cools, the thermistor's resistance increases (1), so the total resistance increases (1) and the current decreases (1).

- 9 A kettle has a metal case connected to the earth wire. Explain why the earth wire is needed, including the role of the fuse. [5 marks]

If the live wire touches the case (1), the earth wire gives a low-resistance path to earth (1); a very large current flows (1), which melts the fuse (1) and disconnects the live wire, so the case is safe to touch (1).

- 10 A 1.5 kW heater runs on the 230 V mains for 4.0 minutes. Calculate (a) the energy transferred and (b) the current. [5 marks]

(a) $P = 1500 \text{ W}$, $t = 240 \text{ s}$ (1); $E = P t = 1500 \times 240$ (1) = 360 000 J (1). (b) $I = P \div V = 1500 \div 230$ (1) = 6.5 A (1).

- 11 Use the equations $P = V I$ and $P = I^2 R$ to explain why the National Grid transmits electricity at a very high p.d. [4 marks]

For a fixed power, $P = V I$, so a higher p.d. means a smaller current (2). $P = I^2 R$, so a smaller current means much less heating in the cables and less energy wasted (2).

- 12 **HIGHER TIER** An ideal step-up transformer has 25 kV and 400 A in the primary coil. The secondary p.d. is 400 kV. Calculate the secondary current. [3 marks]

$$V_p I_p = V_s I_s \text{ (1); } I_s = 25\,000 \times 400 \div 400\,000 \text{ (1)} = 25 \text{ A (1)}$$



REVIEW LATER
FIX-IT LOG

Turn every lost mark into a target.

Q	What went wrong?	The correct physics (or method) is...	Lesson to revisit

EXTENSION: STATIC ELECTRICITY

Only for students taking GCSE Physics (separate science). Combined Science students can skip this page.

When two insulating materials are rubbed together, **electrons** are transferred from one to the other. The one that **gains electrons** becomes **negatively** charged; the one that **loses electrons** is left **positively** charged. Only electrons move, never positive charge.

Like charges repel; unlike charges attract. These are non-contact forces: they act without the objects touching.

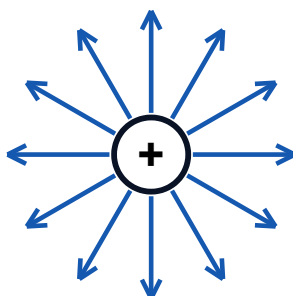
A charged object creates an **electric field** around it. Another charged object in the field feels a force. The field is **strongest close to the object** and gets weaker further away.

If enough charge builds up, the p.d. between the object and the earthed conductor nearby becomes large enough for charge to jump across the air: a **spark**.

- 1 A plastic rod becomes negatively charged when it is rubbed with a cloth. Which way did the electrons move? What charge is left on the cloth? [2 marks]

Electrons moved from the cloth to the rod (1). The cloth is left positively charged (1).

- 2 Draw the electric field around an isolated, positively charged sphere. Add arrows, and show where the field is strongest. [3 marks]



Radial lines pointing outwards; closer together (stronger) near the sphere.

- 3 A small negatively charged sphere is moved towards a large positively charged sphere. Explain what happens to the force on the small sphere, without the spheres touching. [4 marks]

The small sphere is in the electric field of the large sphere (1). Unlike charges attract, so the force is attractive (1), and it is a non-contact force (1). The field is stronger closer to the sphere, so the force increases as they get closer (1).

KEY WORDS

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

Alternating current (ac) · Current that keeps changing direction. UK mains is ac at 50 Hz.

Ammeter · Measures current. Connected in series.

Ampere (A) · Unit of current. $1\text{ A} = 1\text{ C per second}$.

Anomaly · A result that does not fit the pattern of the others.

Battery · Two or more cells joined together.

Charge (Q) · Carried by particles such as electrons. Unit: coulomb (C).

Circuit breaker · A safety switch that cuts off the current if it gets too large.

Conventional current · The agreed direction of current: from + to –.

Current (I) · The rate of flow of charge.

Diode · Lets current flow in one direction only.

Direct current (dc) · Current that flows in one direction only, e.g. from a cell.

Directly proportional · If one quantity doubles, the other doubles. The graph is a straight line through the origin.

Earth wire · Green and yellow safety wire; carries a current only if there is a fault.

Electron · A tiny negatively charged particle. Free electrons carry the current in metals.

Filament lamp · A lamp with a thin metal filament; its resistance increases as it heats up.

Fuse · A thin wire that melts and breaks the circuit if the current is too large.

I–V characteristic · A graph of current against p.d. for a component.

LDR · Light-dependent resistor: resistance decreases as light intensity increases.

LED · Light-emitting diode: a diode that gives out light.

Live wire · Brown wire at about 230 V; dangerous to touch.

National Grid · The system of cables and transformers linking power stations to consumers.

Neutral wire · Blue wire that completes the circuit; at or close to 0 V.

Ohm (Ω) · Unit of resistance.

Ohmic conductor · Current is directly proportional to p.d. at constant temperature.

Parallel · Components on separate branches. A voltmeter is connected in parallel.

Potential difference (V) · Energy transferred per unit charge between two points. Unit: volt (V).

Power (P) · Energy transferred per second. Unit: watt (W).

Resistance (R) · How much a component opposes the current. Unit: ohm (Ω).

Series · Components connected one after another in a single loop.

Thermistor · Resistance decreases as temperature increases.

Transformer · Changes the size of an alternating p.d. (step-up or step-down).

Volt (V) · Unit of p.d. $1\text{ V} = 1\text{ J per C}$.

Voltmeter · Measures p.d. Connected in parallel, across a component.

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN...	R · A · G
11 · Power and energy transfer	
describe power as energy transferred per second	
use $E = P t$ with unit conversions	
use $P = V I$ and $P = I^2 R$	
choose the right equation from the data given	
12 · Mains, safety and the National Grid	
describe the difference between ac and dc	
state the UK mains p.d. and frequency	
name the three wires and their colours and jobs	
explain how the earth wire and fuse protect a user	
explain why the grid uses step-up and step-down transformers	

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

LESSON 11 · POWER AND ENERGY TRANSFER

Do now: (1) $E = Q V = 60 \text{ J}$ (2) $Q = 2.0 \times 60 = 120 \text{ C}$ (3) $R = 230 \div 5.0 = 46 \Omega$

Cover the page: (1) Energy transferred per second. (2) 3500 W (3) $P = V I$ and $P = I^2 R$ (4) $E = P t$

LESSON 12 · MAINS, SAFETY AND THE NATIONAL GRID

Do now: (1) $P = V I = 2300 \text{ W}$ (2) It falls to a quarter. (3) Watt (W).

Cover the page: (1) 230 V, 50 Hz. (2) Brown. (3) Only when there is a fault. (4) Smaller current, so less heating and energy loss in the cables.